Notes: Antràs and Costinot (QJE, 2011)

Intermediated Trade

Contents

1 Big picture

This paper asks what changes when international trade is not conducted directly by producers in frictionless centralized markets, but instead requires **intermediaries**. Standard Ricardian trade theory assumes that goods can be taken to Walrasian markets and exchanged at market-clearing prices. Antràs and Costinot keep the Ricardian core – two islands, two goods, comparative advantage – but insert a realistic market institution: farmers must first match with traders, and that matching process is costly and frictional.

- **Core economic friction:** Farmers cannot directly access centralized markets. They need traders to intermediate trade.
- **Core mechanism:** Search frictions create a lock-in effect. Once a farmer meets a trader, the trader can bargain for a positive margin.
- **Core distinction:** The paper separates two types of integration:
  - W-integration: integration of centralized goods markets, so traders from different islands can exchange goods at common Walrasian prices.
  - M-integration: integration of matching markets, so foreign traders can enter the local producer-intermediary matching market.
- **Main result:** W-integration always produces gains and intermediation magnifies them. M-integration can increase output but may reduce welfare because foreign intermediaries can worsen the bargaining environment and aggravate search externalities.

Interpretation: The paper is not simply saying that "middlemen are good" or "middlemen are bad." It shows that the welfare effect depends on *which market is integrated.* Price integration and intermediary-entry integration have sharply different consequences.

2 Incremental contribution of the paper

2.1 Contribution relative to standard trade theory

The paper adds market institutions to a Ricardian model. In a standard Ricardian model, opening to trade means equalizing goods prices and reallocating production according to comparative advantage. Here, opening to trade also changes the **entry of traders**, the **frequency of successful matches**, and the **terms of exchange between producers and intermediaries**.

2.2 Contribution relative to intermediation and search models

The paper imports search-and-matching logic into international trade. Traders are not passive conduits. They enter when expected intermediation profits justify the cost, and their entry affects producer welfare by changing the probability that unmatched producers find trading opportunities.

2.3 Contribution relative to policy debates about globalization

The paper formalizes a common development concern: foreign trading companies can increase output but still leave producers worse off if bargaining power and search externalities generate an unfavorable equilibrium. The contribution is not an anecdotal claim about exploitation; it is a general-equilibrium mechanism.

2.4 What is new in one sentence

The paper shows that **intermediation magnifies the gains from goods-market integration, but can make intermediary-entry integration welfare-reducing even when it increases output**.

3 Theoretical setting and main objects

3.1 Agents and goods

There are two islands, North and South, and two homogeneous goods, coffee and sugar. Farmers can produce either good, but cannot directly trade in Walrasian markets. Traders do not produce, but have access to Walrasian markets and can intermediate farmers' output.

- Farmers choose what to produce.
- Traders choose whether to become active and pay an intermediation cost τ.
- Farmers and traders meet through a matching function $m(u_F, u_T)$.
- The level of intermediation is $\theta = u_T/u_F$, the ratio of unmatched traders to unmatched farmers.
- Matches are destroyed at Poisson rate λ.

3.2 Search and bargaining

When a farmer and trader meet, they bargain over the surplus. Traders receive a Nash-bargaining share β of ex post gains from trade. Since matching is costly and farmers cannot instantly access another trader, traders earn positive margins.

The paper interprets the trader margin α as the wedge between the Walrasian price and the effective price received by farmers. A larger α means that intermediaries capture more of the joint surplus.

4 Autarky equilibrium

4.1 Ricardian benchmark inside the model

Even with intermediation, the autarkic relative price satisfies

$$p = \frac{a_C}{a_S}.$$

This is the same relative price as in the frictionless Ricardian model. The reason is that search frictions are symmetric across the two goods, so they distort the distribution of surplus between farmers and traders, but not the relative supply decision between coffee and sugar.

4.2 Trader margins and intermediation

The trader's margin is

$$\alpha = \beta - (1 - \beta)(\theta - 1)\frac{\tau}{v(p)}.$$

The equilibrium level of intermediation solves

$$\frac{v(p) - \tau}{\tau} = \frac{r + \lambda + (1 - \beta)\mu_F(\theta)}{\beta\mu_T(\theta)}.$$

Interpretation: Intermediation is higher when the surplus from exchange is larger, when the intermediation cost is lower, or when traders have higher bargaining power. Margins are not fixed markups; they are equilibrium objects shaped by search congestion, bargaining, and free entry.

5 Integration of Walrasian markets: W-integration

5.1 Definition

Under W-integration, traders from both islands can trade in a common centralized goods market, but farmers can still only meet local traders. This is the closest analogue to standard trade liberalization: goods prices converge internationally, but producer access to intermediaries remains domestic.

5.2 Effects on specialization, intermediation, and margins

Because South has comparative advantage in coffee, W-integration raises the relative price of coffee for Southern trader-farmer pairs. Southern farmers specialize in coffee, and the higher surplus from a match induces trader entry. The new equilibrium satisfies

$$\alpha^W = \beta - (1 - \beta)(\theta^W - 1)\frac{\tau}{v(p^W)},$$

$$\frac{v(p^W) - \tau}{\tau} = \frac{r + \lambda + (1 - \beta)\mu_F(\theta^W)}{\beta\mu_T(\theta^W)}.$$

Since $v(p^W) > v(p)$, the level of intermediation rises: $\theta^W > \theta$.

5.3 Magnification effect

The paper's first big welfare result is that intermediation **magnifies** gains from trade under W-integration. The intuition is:

- Standard Ricardian gains arise because South sells coffee at a better relative price.
- Additional gains arise because the higher surplus attracts more traders.
- More traders increase the probability that farmers are matched.
- More matching means more of the economy can actually realize the potential gains from trade.

Economic message: In a frictionless model, price integration only changes relative prices and production. In this model, price integration also improves market access by increasing intermediary entry. That is why the gains are larger than in the benchmark Ricardian model.

6 Integration of matching markets: *M*-integration

6.1 Definition

Under M-integration, foreign traders can enter the local matching market and search for local farmers. This is closer to a form of foreign direct investment or entry by foreign trading companies. The paper assumes Northern traders may be more efficient and may also have higher bargaining power.

6.2 The key lemma

Once M-integration occurs, new matches involve Northern traders with probability one. Southern traders are displaced from the matching market because Northern traders are more profitable: they have lower intermediation costs and/or stronger bargaining positions.

6.3 Output effects versus welfare effects

M-integration always increases the level of intermediation and therefore increases output along the transition path. But welfare is ambiguous. The reason is that a higher matching rate helps farmers, while higher bargaining power of foreign traders can reduce the share of surplus farmers receive.

The change in unmatched Southern farmers' expected lifetime utility can be written as

$$\Delta V_F^U = \frac{\theta^N \tau^*(1-\bar{\beta})}{r\bar{\beta}} - \frac{\theta^W \tau(1-\beta)}{r\beta}.$$

If the second term dominates, unmatched Southern farmers lose from M-integration. Since changes in social welfare are tied to ΔV_F^U, aggregate welfare can fall.

6.4 Economic mechanism: the search externality

The source of possible losses is not a simple transfer from South to North. Instead, it is a search externality. Bargaining between a farmer and a trader affects not only their own surplus split, but also trader entry and the matching probabilities faced by all unmatched farmers and traders. Bilateral bargaining does not internalize these economy-wide effects.

Interpretation: This generates a prisoner's dilemma. Each individual farmer is willing to trade with a Northern intermediary once matched, but if all farmers do so, Southern intermediaries exit and the bargaining environment can become worse for farmers as a group.

7 Extensions and robustness

The paper checks whether the core results depend on restrictive assumptions.

- **Large Northern traders:** Large foreign trading companies can microfound higher bargaining power. Welfare losses can be interpreted as arising from inefficiently large intermediary organizations.
- **Endogenous farmers, exogenous traders:** If traders are fixed by regulations and farmers choose whether to participate in markets, W-integration still raises welfare, while M-integration can still create losses.
- **Occupational choice:** If agents can choose whether to be farmers or traders, W-integration still makes agents weakly better off, while M-integration can still generate winners, losers, and aggregate welfare losses.
- **Slow reallocation or crop-specific traders:** If farmers cannot instantly switch goods or traders are sector-specific, W-integration may create richer distributional effects, but aggregate gains remain.
- **Heterogeneous farmers:** Productive farmers may bypass intermediaries and access markets directly, while less productive farmers rely on traders. This aligns with evidence that less productive exporters tend to use intermediaries.

8 Original extracted model and result panels

The article is a theory paper and contains no formal numbered empirical tables or data figures. To follow the same visual-note standard, the panels below directly extract the original equation, lemma, and proposition blocks from the paper. Every two extracted panels are placed on the same row.

8.1 Visual Panels 1–2: Basic Environment and Trader Utility / Intermediation Technology

II. THE BASIC ENVIRONMENT

Consider an island inhabited by a continuum of infinitely lived agents consuming two goods, coffee (C) and sugar (S). An exogenous measure N_F of the island inhabitants are engaged in production.[10] We refer to this set of agents as *farmers* and assume that they (and only they) have access to production technologies that allow them to produce an amount $1/a_C$ of coffee or an amount $1/a_S$ of sugar per unit of time. A farmer cannot produce both goods at the same date t and goods are not storable. We denote by $\gamma \in [0, 1]$ the share of coffee farmers at a given date. For notational convenience, we drop time indices from all our variables whenever there is no risk of confusion.

Our main point of departure from the classical Ricardian model is that farmers do not have direct access to Walrasian markets where their output can be exchanged for that of other farmers. To be able to sell part of their output and consume both goods,

Panel 1: Basic environment - farmers, traders, coffee, and sugar

The pool of potential traders on the island is large. At any point in time, potential traders can become active or inactive. To remain connected to Walrasian markets, an active trader must incur an intermediation cost equal to τ at each date, but stands to obtain some remuneration when intermediating a trade for a farmer. By contrast, inactive traders are involved in an activity that generates no income but also no disutility of effort, for example, laying in a hammock.[12] We assume that the pool of potential traders is large enough to ensure that the measure of traders operating on the island, N_T, is not constrained by population size and some agents are always laying in hammocks. Hence, in equilibrium, N_T will be endogenously pinned down by free entry.

All agents aim to maximize the expected value of their lifetime utility[13]

$$V = E\left[\int_0^{+\infty} e^{-rt}\left[v\left(C(t), S(t)\right) - \mathcal{I}_A(t)\,\tau\right] dt\right],$$

where $r > 0$ is the common discount factor; $\mathcal{I}_A(t) = 1$ if the agent is an active trader at date t and $\mathcal{I}_A(t) = 0$ otherwise; $C(t) \geq 0$ and $S(t) \geq 0$ are the consumption of good C and S at date t, respectively; and v is increasing, concave, homogeneous of degree 1 and satisfies the two Inada conditions: $\lim_{C\to 0} v_C = \lim_{S\to 0} v_S = +\infty$ and $\lim_{C\to+\infty} v_C = \lim_{S\to+\infty} v_S = 0$. The assumption that the utility

Panel 2: Trader entry, intermediation cost, and lifetime utility

Visual Panels 1–2: Basic Environment and Trader Utility / Intermediation Technology

Read: Panel 1 introduces the core departure from standard Ricardian trade: farmers can produce, but they need traders to access markets. Panel 2 formalizes the trader side: active traders pay an intermediation cost and agents maximize discounted expected utility.

Economic message: The model turns market access into an equilibrium outcome. Trade gains depend not only on comparative advantage, but also on how easily producers find intermediaries.

8.2 Visual Panels 3–4: Bellman Equations (1)–(4) and Nash Bargaining Equations (5)–(7)

for $i = C, S$, are determined. These value functions must satisfy the following Bellman equations:

(1) $rV_F^U = \mu_F(\theta)\left[\max\left\{V_{F_C}^M, V_{F_S}^M\right\} - V_F^U\right] + \dot{V}_F^U,$

(2) $rV_{F_i}^M = v(C_{F_i}, S_{F_i}) + \lambda\left(V_F^U - V_{F_i}^M\right) + \dot{V}_{F_i}^M,$

(3) $rV_T^U = -\tau + \mu_T(\theta)\left[\gamma\left(V_{T_C}^M - V_T^U\right) + (1-\gamma)\left(V_{T_S}^M - V_T^U\right)\right] + \dot{V}$

(4) $rV_{T_i}^M = v(C_{T_i}, S_{T_i}) - \tau + \lambda\left(V_T^U - V_{T_i}^M\right) + \dot{V}_{T_i}^M.$

Panel 3: Bellman equations for farmers and traders, equations (1)–(4)

We can now describe how the process of intermediation and Nash bargaining between farmers and traders affect the division of surplus and the implied terms of exchange of goods C and S. As we formally show in the Appendix, Nash bargaining between farmers and traders implies that at any point in time,

(5) $$V_{T_i}^M - V_T^U = \beta\left(V_{T_i}^M + V_{F_i}^M - V_T^U - V_F^U\right)$$

as well as

(6) $$\frac{v_C(C_{F_i}, S_{F_i})}{v_S(C_{F_i}, S_{F_i})} = \frac{v_C(C_{T_i}, S_{T_i})}{v_S(C_{T_i}, S_{T_i})} = p$$

and

(7) $$p\bar{C}_i + \bar{S}_i = (p/a_C)\cdot\mathcal{I}_C + (1/a_S)\left(1 - \mathcal{I}_C\right),$$

where $\bar{C}_i \equiv C_{F_i} + C_{T_i}$ and $\bar{S}_i \equiv S_{F_i} + S_{T_i}$ denote the joint consumption of coffee and sugar by each farmer-trader match producing good $i = C, S$, respectively. Equation 5 simply states that traders get a share β of the surplus of any match, whereas Equations 6

Panel 4: Nash bargaining and efficient matched-pair consumption, equations (5)–(7)

Visual Panels 3–4: Bellman Equations (1)–(4) and Nash Bargaining Equations (5)–(7)

Read: Panel 3 writes the values of matched and unmatched farmers and traders. Panel 4 shows how bilateral Nash bargaining splits the match surplus while preserving efficient consumption within each matched pair.

Economic message: The wedge in this model is not technological inefficiency inside a match. It is the distributional and entry distortion generated by costly search and bargaining.

8.3 Visual Panels 5–6: Trader Margin Equation (17) and Intermediation Equations (18)–(20)

The joint instantaneous utility enjoyed by a matched farmer-trader pair is thus given by $v(\bar{C}, \bar{S}) - \tau$ and is time-invariant. Because the function $v(\cdot)$ is homogeneous of degree 1, it is also necessarily the case that $v(\cdot)$ is proportional to the value of the farmer's good in the Walrasian market (i.e., the joint spending of the matched pair). In the rest of the paper, we slightly abuse notation and denote by $v(p) \equiv v(\bar{C}, \bar{S})$ the joint utility level (net of effort costs) of a matched farmer-trader pair when the relative price of coffee is equal to p.

We next turn to a discussion of the terms of trade in bilateral exchanges, which is at the heart of our analysis. Throughout the paper, we denote by $\alpha \in (0,1)$ the share of joint consumption $\bar{C}$ and $\bar{S}$ that is captured by the trader, with the remaining share $1-\alpha$ accruing to the farmer. Equation 6 ensures that this share is common for both goods. Naturally, a higher α is associated with a distribution of surplus that is more favorable to the trader. As shown in the Appendix, Equations 1–5 imply that at all points in time in the autarky equilibrium, the share α is given by

$$\alpha = \beta - \frac{(1-\beta)(\theta-1)\tau}{v(p)}. \tag{17}$$

Not surprisingly, the previous expression states that the share α of goods captured by the trader is decreasing in the ratio θ of unmatched traders to unmatched farmers. Straightforward manipulation of Equation 17 also demonstrates that for a given value of θ, α is necessarily increasing in the primitive bargaining power β.

The value of α can be interpreted as the traders' margins, that is, the (percentage) difference between the world relative

Panel 5: Trader margin, equation (17)

Having discussed the determination of prices in our model, we next move to characterizing the dynamics of the level of intermediation, the value functions, and the measures of matched and unmatched traders and farmers on the island. Using the free entry condition 10, which of course implies $\dot{V}_T^U = 0$, we can rearrange Equation 3 as

$$V_T^M = \frac{\tau}{\mu_T(\theta)}. \tag{18}$$

Equation 18 simply states that the present discounted utility of a matched trader should be equal to the present discounted utility cost of remaining active while searching for a match. It implicitly defines the level of intermediation θ as an increasing function of the value function V_T^M, $\theta \equiv \hat{\theta}(V_T^M)$. To characterize the dynamics of the level of intermediation, we can therefore focus on the dynamics of V_T^M. Combining the Bellman equation of matched traders 4 with the free entry condition 10 and the Nash bargaining outcome 17, we obtain

$$\dot{V}_T^M = (r+\lambda)V_T^M + (1-\beta)\hat{\theta}(V_T^M)\tau - \beta[v(p)-\tau]. \tag{19}$$

Since we know that $\hat{\theta}'(V_T^M) > 0$ by 18, we can conclude that the dynamics of V_T^M in Equation 19 are unstable. For the expected lifetime utility of a matched trader to remain finite we therefore need $\dot{V}_T^M = 0$, which further implies $\dot{\theta} = \dot{\alpha} = 0$. Using the fact that $\dot{V}_T^M = 0$ with Equations 18 and 19, the equilibrium level of intermediation θ can then be expressed, *at any point in time*, as the implicit solution of

$$\frac{v(p)-\tau}{\tau} = \frac{r+\lambda+(1-\beta)\mu_F(\theta)}{\beta\mu_T(\theta)}. \tag{20}$$

Panel 6: Free entry and equilibrium intermediation, equations (18)–(20)

Visual Panels 5–6: Trader Margin Equation (17) and Intermediation Equations (18)–(20)

Read: Panel 5 defines the trader margin α. Panel 6 links the value of being a matched trader to free entry and pins down θ, the equilibrium level of intermediation.

Economic message: A trader's margin is endogenous. Higher trader entry improves farmers' outside options and tends to lower margins, while higher bargaining power raises margins.

8.4 Visual Panels 7–8: Proposition 1 - Autarkic Equilibrium and W-integration Equations (26)–(27)

As shown in the Appendix, the right-hand side of this last equation is increasing in θ and hence, the steady-state measure of traders N_T is higher in economies with better production technologies, lower intermediation costs τ, and higher bargaining power β of traders.

The previous discussion has demonstrated, by construction, the existence and uniqueness of an autarkic equilibrium. It has also characterized some of its key features, as summarized in Proposition 1.

PROPOSITION 1. An autarkic equilibrium exists and is unique. The relative price of coffee, p, the share of coffee farmers, γ, the vector of consumption levels, (C_F, S_F, C_T, S_T), and the level of intermediation, θ, are constant over time and determined by Equations 13–17 and 20. Similarly, the lifetime utilities of all agents are time-invariant and given by Equations 10, 18, 21, and 22. By contrast, the measures of matched and unmatched farmers and traders slowly converge to their steady-state value, Equations 23–25.

Panel 7: Proposition 1 - autarkic equilibrium

Since Southern farmers can only match with traders from their own island, we can use the same argument as in Section III to show that the traders' margins, α^W, and the level of intermediation, θ^W, will immediately jump to their new steady-state values given by:

$$\alpha^W = \beta - \frac{(1-\beta)\left(\theta^W - 1\right)\tau}{v\left(p^W\right)}, \tag{26}$$

$$\frac{v\left(p^W\right) - \tau}{\tau} = \frac{r + \lambda + (1-\beta)\,\mu_F\left(\theta^W\right)}{\beta\mu_T\left(\theta^W\right)}. \tag{27}$$

Equations 26 and 27 are just the counterparts of 17 and 20 with $v\left(p^W\right) > v(p)$. Using the two previous expressions, all other equilibrium variables can then be computed by simple substitutions. In particular, all value functions must directly jump to their new steady-state values after W-integration.[18]

Panel 8: W-integration margins and intermediation, equations (26)–(27)

Visual Panels 7–8: Proposition 1 - Autarkic Equilibrium and W-integration Equations (26)–(27)

Read: Proposition 1 summarizes existence, uniqueness, prices, margins, intermediation, and value functions under autarky. Panel 8 shows how the same logic applies after Walrasian markets are integrated.

Economic message: The model is a strict generalization of Ricardian trade. When market access becomes frictionless, the model converges back toward the standard benchmark.

8.5 Visual Panels 9–10: W-integration Distributional Equation (28), Magnification Effect, and Proposition 2

First, the instantaneous increase in θ will slowly increase the number of matched farmers in the economy, as illustrated by Equation 11. Starting from the autarky equilibrium, W-integration therefore leads to GDP growth along the transition path toward the new steady-state equilibrium.[19] The magnitude of this "growth effect" depends on the initial level of intermediation as well as the properties of the matching technology. If the matching elasticity $\varepsilon \equiv \frac{d \ln m(u_F, u_T)}{d \ln u_T}$ is nonincreasing in the level of intermediation, then ceteris paribus, islands with lower levels of intermediation always grow faster after W-integration (see Appendix).[20] In this situation, trade integration tends to lead to convergence across countries.

Second, the endogenous increase in the level of intermediation due to W-integration has distributional consequences. Combining Equations 26 and 27, we get

$$\alpha^W = \beta \cdot \left[\frac{r + \lambda + \mu_T \left(\theta^W\right)}{r + \lambda + (1-\beta)\, \mu_F \left(\theta^W\right) + \beta \mu_T \left(\theta^W\right)} \right], \tag{28}$$

where the bracket term is decreasing in θ^W. Thus the instantaneous entry of new traders reduces α^W, and this implies an instantaneous improvement of the farmers' terms of trade and an instantaneous worsening of the traders' terms of trade.

Panel 9: W-integration distributional equation (28)

where

$$\Omega(t) \equiv \frac{r\left[N_F - u_F(t)\right] + (1-\beta)\, \mu_F\left[\theta(t)\right] N_F}{r + \lambda + \beta \mu_T \left[\theta(t)\right] + (1-\beta)\, \mu_F \left[\theta(t)\right]}.$$

As explained above, W-integration raises the surplus from trading, as captured by the utility term $v\left[p(t)\right]$. This is the standard welfare gain highlighted by neoclassical models of trade. Notice, however, that $\Omega(t)$ is increasing in the level of intermediation $\theta(t)$ and hence it also increases following W-integration. We can then conclude that compared with a standard Ricardian model, in which $\tau = 0$ and so $\theta(t) = +\infty$, the integration of Walrasian markets leads to a higher (percentage) increase in social welfare. We refer to this result as the "magnification effect" of intermediation. This is, of course, the welfare counterpart of the growth effect discussed in the previous section.[21]

Proposition 2 summarizes our findings about the effects of W-integration.

PROPOSITION 2. W-integration: (i) induces growth along the transition path and, if the matching elasticity ε is nonincreasing in the level of intermediation, leads to convergence across islands; (ii) improves the farmers' terms of trade and worsens the traders' terms of trade; and (iii) makes all agents (weakly) better off.

In the case where the Southern island is not small relative to

Panel 10: W-integration magnification effect and Proposition 2

Visual Panels 9–10: W-integration Distributional Equation (28), Magnification Effect, and Proposition 2

Read: Panel 9 shows that the new entry of traders under W-integration lowers trader margins and improves farmers' terms of trade. Panel 10 states Proposition 2: W-integration induces growth, improves farmers' terms of trade, worsens traders' terms of trade, and makes all agents weakly better off.

Economic message: Price integration is unambiguously good in the model because it raises trade surplus and simultaneously deepens the intermediary market.

8.6 Visual Panels 11–12: Lemma 1 - Northern Trader Entry and M-integration Equations (36)–(38)

power. Since the surplus being generated by a match with a Northern trader is higher, $v\left(p^W\right) - \tau^* > v\left(p^W\right) - \tau$, farmers are also more likely to stay in a match that involves a Northern trader than to keep searching for another type of trader. Putting all the previous pieces together, we have that Northern traders will necessarily be more profitable (i.e., attain higher welfare levels) than Southern traders under random matching; see Appendix for details. Appealing to free entry, we then obtain the following lemma.

LEMMA 1. If M-integration occurs at some unexpected date t_0, then with probability 1, new matches only involve Northern traders in both islands for all $t > t_0$.

Panel 11: Lemma 1 - new matches involve Northern traders

Combining the previous expression with Equations 30–33 and our Nash bargaining conditions, it is easy to verify that the Northern traders' margins, α^N, and the level of intermediation after M-integration, θ^N, will immediately satisfy

$$\alpha^N = \bar{\beta} - \frac{\left(1-\bar{\beta}\right)\left(\theta^N - 1\right)\tau^*}{v\left(p^W\right)}, \tag{36}$$

$$\frac{v\left(p^W\right) - \tau^*}{\tau^*} = \frac{r + \lambda + \left(1-\bar{\beta}\right)\mu_F\left(\theta^N\right)}{\bar{\beta}\mu_T\left(\theta^N\right)}. \tag{37}$$

unmatched Northern traders in South is exactly equal to 0 (independently of the measure of Southern farmers searching for matches).

Combining Equations 34 and 35 with our Nash bargaining conditions, we can also show (see proof of Lemma 1 for details) that the margins of Southern traders must also immediately jump to

$$\alpha^S = \beta - \frac{(1-\beta)\left[\xi\theta^N - 1\right]\tau}{v\left(p^W\right)}, \tag{38}$$

Panel 12: M-integration margins and intermediation, equations (36)–(38)

Visual Panels 11–12: Lemma 1 - Northern Trader Entry and M-integration Equations (36)–(38)

Read: Lemma 1 states that after matching-market integration, new matches involve Northern traders. Panel 12 shows how the post-integration margins and intermediation level are governed by Northern trader characteristics.

Economic message: Foreign intermediary entry improves matching technology but changes bargaining power. This is why output and welfare can move in different directions.

8.7 Visual Panels 13–14: M-integration Value Functions (39)–(42) and Welfare Equation (44)

(39) $$V_F^U = \frac{\mu_F\left(\theta^N\right)\left(1-\alpha^N\right) v\left(p^W\right)}{r\left[r+\lambda+\mu_F\left(\theta^N\right)\right]},$$

(40) $$V_{F^N}^M = \frac{\left(1-\alpha^N\right) v\left(p^W\right)+\lambda V_F^U}{r+\lambda},$$

(41) $$V_{F^S}^M = \frac{\left(1-\alpha^S\right) v\left(p^W\right)+\lambda V_F^U}{r+\lambda},$$

(42) $$V_{T^S}^M = \frac{\alpha^S v\left(p^W\right)-\tau}{r+\lambda}.$$

Panel 13: M-integration value functions, equations (39)–(42)

any date t before M-integration and after W-integration, we know that

$$W(t) = V_F^U(t)\left[u_F(t)+\frac{\lambda\left[N_F-u_F(t)\right]}{r+\lambda}\right]+\left[v\left(p^W\right)-\tau\right]\left[\frac{N_F-u_F(t)}{r+\lambda}\right].$$

Since $v\left(p^W\right)$ is not affected by M-integration and $u_F(t)$ is predetermined at date t, the previous expression implies that changes in social welfare caused by M-integration, ΔW, must reflect changes in the expected lifetime utility of unmatched farmers, ΔV_F^U. Given our earlier discussion of the relationship between V_F^U, $V_{T^S}^M$, and α^S, this further implies the Southern traders' terms of trade, α^S, is a sufficient statistic for welfare analysis in the South.[30] In particular, there will be aggregate losses from M-integration in the South, $\Delta W<0$, if and only if $\alpha^S>\alpha^W$.[31]

Using Equations 36, 37, and 39 as well as their counterparts under W-integration, we can compute explicitly the change in the expected lifetime utility of unmatched farmers caused by M-integration as

(44) $$\Delta V_F^U = \frac{\theta^N\tau^*(1-\bar{\beta})}{r\bar{\beta}}-\frac{\theta^W\tau(1-\beta)}{r\beta}.$$

As our analysis of the distributional consequences of M-integration already anticipates, it will prove useful to separate the rest of our welfare analysis into two parts. First, we consider the case in which differences in intermediation costs are the only difference in market institutions across the two islands: $\tau<\tau^*$, but $\beta\to\bar{\beta}$. Second, we turn to the polar case in which intermediation costs are similar across countries, $\tau\to\tau^*$, but bargaining powers are

Panel 14: M-integration welfare condition, equation (44)

Visual Panels 13–14: M-integration Value Functions (39)–(42) and Welfare Equation (44)

Read: Panel 13 gives the value functions after Northern trader entry. Panel 14 shows that the sign of the welfare effect depends on the change in unmatched Southern farmers' expected utility.

Economic message: The sufficient statistic for welfare is not simply output or entry. It is how integration changes the outside option and surplus share of local producers.

8.8 Visual Panels 15–16: Proposition 3 - M-integration and Proposition 4 - Large Northern Traders

because Northern traders have an even higher bargaining power, and thus social welfare is driven down. This result clearly echoes Bhagwati's (1971) celebrated results on trade and domestic distortions.

An obvious question at this point is: if unmatched Southern farmers are worse off under M-integration, why do they trade with Northern traders? The answer is that random matching—which we believe fittingly captures search frictions in an environment where traders are mobile, but farmers are not—leads to a simple prisoner's dilemma situation. Although all Southern farmers are worse off in the equilibrium in which only Northern traders are active, each Southern farmer individually has an incentive to trade with Northern traders. This is true both ex ante and ex post, that is, both before and after matches occur. Even if Southern farmers had the choice to commit not to trade with Northern traders ex ante, each farmer would strictly prefer to trade with Northern traders, independently of what other traders are doing. The intuition is the following. Because of Nash bargaining, Northern traders always give Southern farmers more than what they would get if unmatched. Because farmers are all of measure 0, they do not internalize the impact of their own actions on the composition of traders in the island. As a result, farmers are always better off trading with Northern traders, thereby leading to the exit of all unmatched Southern traders. If the primitive bargaining power of Northern traders is high enough, this may lead to lower aggregate welfare in the Southern island (and the world as a whole).

Our main results about the impact of M-integration are summarized in the next proposition.

PROPOSITION 3. M-integration: (i) always induces growth along the transition path and, if the matching elasticity ε is nonincreasing in the level of intermediation, leads to convergence across islands; (ii) always creates winners and losers; and (iii) may decrease aggregate welfare.

Panel 15: Proposition 3 - M-integration growth, winners/losers, and possible welfare losses

$$rV_F^U = \mu_F\left(\theta^N\right)\left[\phi\left(V_{F^N}^M - V_F^U\right) + (1-\phi)\left(V_{F^S}^M - V_F^U\right)\right] + \dot{V}_F^U,$$

$$r\underline{V}_F^U = \mu_F\left(\theta^N\right)\left[\left(\frac{n-1}{n}\right)\phi\left(V_{F^N}^M - \underline{V}_F^U\right) + (1-\phi)\left(V_{F^S}^M - \underline{V}_F^U\right)\right] + \dot{\underline{V}}_F^U$$

where $\phi \equiv u_{T^N}/[u_{T^N} + u_{T^S}]$ is the share of unmatched Northern traders, and $V_{F^N}^M \equiv \max\left\{V_{F_C^N}^M, V_{F_S^N}^M\right\}$ and $V_{F^S}^M \equiv \max\left\{V_{F_C^S}^M, V_{F_S^S}^M\right\}$ still denote the expected lifetime utilities of Southern farmers matched with Northern and Southern traders, respectively. The rest of our model is as described in Section V, with x being endogenously determined through free entry.

In this environment, one can show the following proposition.

PROPOSITION 4. The equilibrium under M-integration with large Northern traders is isomorphic to an equilibrium under M-integration with infinitesimally small Northern traders with primitive bargaining power

$$\beta^L \equiv \overline{\beta}\frac{r + \mu_F\left(\theta^N\right)}{r + \mu_F\left(\theta^N\right)\left[\overline{\beta} + \left(\frac{n-1}{n}\right)\left(1-\overline{\beta}\right)\right]} > \overline{\beta}.$$

Panel 16: Proposition 4 - large Northern traders and bargaining power

Visual Panels 15–16: Proposition 3 - M-integration and Proposition 4 - Large Northern Traders

Read: Proposition 3 summarizes the paper's most distinctive result: M-integration induces growth but can reduce aggregate welfare. Proposition 4 shows that large Northern trading companies provide a microfoundation for stronger bargaining power.

Economic message: The paper separates productivity/output gains from welfare gains. A policy can raise output and still hurt local welfare if it worsens the surplus split enough.

8.9 Visual Panels 17–18: Proposition 5 - Endogenous Farmers/Exogenous Traders and Proposition 6 - Occupational Choice

Using the previous equilibrium conditions and the same strategy as in Sections IV and V, one can establish the following proposition.

PROPOSITION 5. Suppose that there is an endogenous number of farmers and an exogenous number of traders. Then W-integration worsens the farmers' terms of trade, improves the traders' terms of trade, and makes all agents (weakly) better off. By contrast, M-integration always creates winners and losers and may decrease aggregate welfare.

Panel 17: Proposition 5 - endogenous farmers and exogenous traders

$$V_T^U \geq V_F^U, \text{ if } N_T > 0,$$

$$V_F^U \geq V_T^U, \text{ if } N_F > 0.$$

This is the counterpart of the free entry condition, Equation (10), in our original model. In any non-degenerate equilibrium with both types of agents being active, the previous conditions, of course, imply that agents must be indifferent between becoming a farmer or a trader: $V_T^U = V_F^U$.

In such an economy, the impact of W- and M-integration can be summarized as follows.

PROPOSITION 6. Suppose that all agents can become farmers or traders at any point in time. Then W-integration does not affect the farmers' and traders' terms of trade and makes all agents (weakly) better off. By contrast, M-integration may create winners and losers and decrease aggregate welfare.

Panel 18: Proposition 6 - occupational choice

Visual Panels 17–18: Proposition 5 - Endogenous Farmers/Exogenous Traders and Proposition 6 - Occupational Choice

Read: Proposition 5 shows that the main contrast between W-integration and M-integration survives when the number of farmers is endogenous and traders are exogenous. Proposition 6 shows that the core result also survives when agents can choose occupations.

Economic message: The paper's mechanism is robust. The possibility of losses from intermediary-entry integration is not an artifact of one narrow occupational structure.

9 Short summary of the paper

- The model starts with a Ricardian economy but adds costly search for intermediaries.
- Intermediaries earn margins because farmers cannot instantly access Walrasian markets.
- W-integration raises the surplus from specializing according to comparative advantage and attracts more intermediaries.
- Therefore, W-integration generates Pareto gains and magnifies the standard gains from trade.
- M-integration allows more efficient or more powerful foreign traders to enter local matching markets.
- M-integration raises intermediation and output, but can lower welfare if foreign traders' higher bargaining power worsens local producers' outside options.
- The welfare losses arise from a search externality and a prisoner's dilemma among producers, not from a simple transfer of rents from South to North.

10 Conclusion

10.1 Core conclusion

The central conclusion is that international trade with intermediaries is not equivalent to international trade in frictionless markets. Intermediaries help producers realize gains from trade, but the market structure through which intermediaries operate matters for welfare.

10.2 Economic intuition

The model distinguishes two layers of globalization. The first layer is goods-price integration. This raises the value of specialization and improves matching because it attracts traders. The second layer is integration of the intermediary market. This can import better trading technology, but it can also import stronger bargaining power and alter the distribution of surplus.

10.3 Incremental contribution revisited

The paper's contribution is to show that intermediation is a double-edged institution. It can enlarge trade gains by making markets work, but when matching markets themselves are liberalized, the change in bargaining structure can generate aggregate welfare losses.

10.4 Implications

For policy, the paper suggests that facilitating access to world prices is not the same as allowing unrestricted entry of foreign intermediaries. If foreign intermediaries enter with much greater bargaining power, policy may need to focus on market design, producer organizations, entry rules, or mechanisms that improve farmers' outside options.

10.5 Limitations

The model is stylized. It abstracts from many real-world details such as multi-layer supply chains, credit constraints, quality certification, risk aversion, and directed search. The paper acknowledges this and treats the framework as a tractable way to isolate the logic of intermediation rather than as a complete description of any particular market.

10.6 Final takeaway

Takeaway: Intermediaries are neither merely rent extractors nor merely facilitators. Their welfare role depends on how trade integration changes prices, matching probabilities, bargaining power, and entry. Under goods-market integration they magnify gains; under matching-market integration they can create both growth and welfare losses.